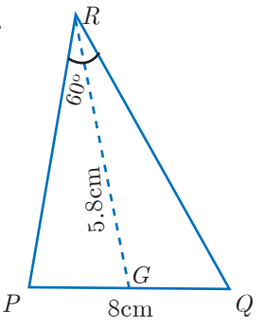
Construction of a triangle when its base, the vertical angle and the median from the vertex of the base are given.

Example 4.17 Construct a ΔPQR in which $PQ = 8$ cm, $\angle R = 60°$ and the median RG from R to PQ is 5.8 cm. Find the length of the altitude from R to PQ.



Rough diagram

Solution

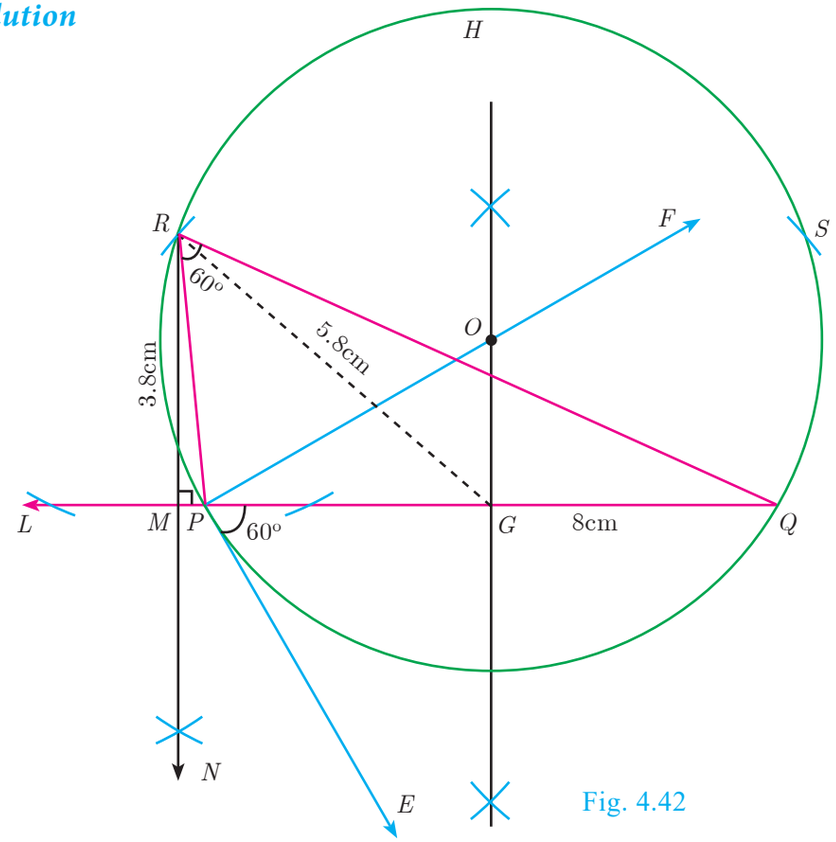


Fig. 4.42

Construction

Step 1: Draw a line segment $PQ = 8$cm.

Step 2: At P, draw PE such that $\angle QPE = 60°$.

Step3: At P, draw PF such that $\angle EPF = 90°$.

Step 4: Draw the perpendicular bisector to PQ, which intersects PF at O and PQ at G.

Step 5: With O as centre and OP as radius draw a circle.

Step 6: From G mark arcs of radius 5.8 cm on the circle. Mark them as R and S.

Step 7 : Join PR and RQ. Then ΔPQR is the required triangle .

Step 8 : From R draw a line RN perpendicular to LQ. LQ meets RN at M

Step 9: The length of the altitude is $RM = 3.8$ cm.

Note

We can get another ΔPQS for the given measurements.

10th_Maths_Chapter 4_English.indd 179 12/27/2024 3:11:14 PM